

PRISMA 2020 checklist

Supplementary Material for the article “Deep Learning for Leaf Disease Recognition in Rice and Maize: A Systematic Review of Vision Tasks and an Audit of Data Leakage in Model Evaluation”. The item texts are reproduced verbatim from Page et al. [1] under a CC BY 4.0 license. Section, table, figure, equation, and page numbers refer to the main manuscript; “data repository” is the data archive on Zenodo, <https://doi.org/10.5281/zenodo.23151397>; “Supplementary Material” is the Supplementary Material file in that archive. The Data Availability Statement and the Declarations section are unnumbered and are located on p. 43. Sentences in the abstract are numbered in order of appearance.

PRISMA 2020 checklist

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
Title				
Title	1	Identify the report as a systematic review.	Title, p. 1	The title states “systematic review”.
Abstract				
Abstract	2	See the PRISMA 2020 for Abstracts checklist (table 2).	Abstract, p. 1	See the abstract checklist below.
Introduction				
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section 1, p. 2; related reviews, p. 2; Table 1, p. 3	Motivation, prior reviews, and three gaps.
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Research questions, p. 4	RQ1–RQ4.
Methods				
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2.1, p. 5; Section 2.2, p. 6; Supplementary Material, Table S2	Inclusion criteria and the four full-text assessment gates; scope amendments and post-registration changes in Table S2 of the Supplementary Material; grouping by task (CLS, DET, SEG, GEN) and crop.
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2.1, p. 5	Scopus (2026-09-21), IEEE Xplore, ScienceDirect, and MDPI (2026-09-22); reference lists, Google Scholar, and other sources were not searched.
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section 2.1, p. 5; Table 2, p. 6	Keyword blocks and verbatim strings, search fields, filters (year, document type, language), run dates, and number of results for all four sources, presented in the main paper.
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.1, p. 5; Figure 1, p. 7; Supplementary Material, Section S2	Round 1: predefined rules on metadata. Round 2: an AI tool (Google Gemini) provided preliminary suggestions; one author reviewed each suggestion, a second author checked the excluded records, and a third author decided disagreements. Full text: one author assessed against the four gates, and another author rechecked every decision. No independent dual screening and no agreement coefficient computed.
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2.2, p. 6 (Data extraction); Supplementary Material, Sections S2 and S4; Declarations, p. 43	One author extracted data using a predefined form, and another author cross-checked the extracted data (not independently); authors of the primary studies were not contacted. An AI tool supported the re-examination of the leakage labels of 92 studies, and one author confirmed each label.

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2.2, p. 6; Section 6.5.3, p. 34	One primary metric for the proposed model on the study's own test set (accuracy, F1, mAP, IoU, Dice, FID); for external validation, a predefined rule selects one pair of values per study.
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2.2, p. 6; data repository	Task, crop, data, model, protocol; information not reported was recorded as not determinable, without inference; field dictionary in the repository.
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.2, p. 6 (Protocol assessment); Equation (2); Supplementary Material, Section S4 and Table S4	Purpose-built tool: the independent-split condition, six leakage signs (TEST-AUG recorded only as a flag), and three levels: clean, suspected leakage, and not determinable; a dataset-level rule for the SETH dataset [2] (SUB-UNIT sign) based on the measured duplicate-image rate (Table 5, p. 15). One author assessed each study and another rechecked every label; 92 labels re-examined with AI-tool support; no agreement coefficient.
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Section 2.2, p. 7; Supplementary Material, Section S6; Equation (10)	Proportions with 95% Wilson CIs; medians and quartiles; Mann–Whitney with r_{tb} ; Hodges–Lehmann; odds ratios; external-validation drop Δ .
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Section 2.2, p. 8 (Selection of representative studies); Figure 2, p. 8	All counts computed over the 202 studies; 38 representative studies by quota; two dataset-specific tables.
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2.2, p. 6; Section 6.5.3, p. 34	Missing information recorded as not determinable; one primary metric per study; rule for selecting the external-validation pair of values.
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2.2, p. 6; Section 6, p. 20	Tables of representative studies, dataset-specific tables, and figures by year, architecture family, data source, imaging conditions, and external-validation drop.
	13d	Describe any methods used to synthesise results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2.2, p. 7 (Statistical methods)	Descriptive synthesis, with no pooling of metrics because data, class sets, and protocols differ; one primary test (two-sided Mann–Whitney); Python, SciPy, and statsmodels; analysis code available upon request.
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section 2.2, p. 7	Comparisons by protocol level, within each data source, by imaging conditions, and by type of shift under external validation.
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesised results.	Section 2.2, p. 7 (Statistical methods)	Van Elteren within each source, rank regression with source fixed effects (without them: descriptive), and Fisher for the proportion of results of 99.5% or higher (three analyses with Holm correction); the three-group comparison including reports excluded for leakage is exploratory.

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Section 11, pp. 41–42	Not assessed. The result of each study is an author-reported accuracy on different data and protocols, so the review does not pool effects; instead, the evaluation protocol of each study is appraised. This limitation, including the risk that only the best results are published, is stated in Section 11, pp. 41–42.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Section 11, pp. 41–42	No GRADE certainty rating because there is no pooled estimate; the confidence in each conclusion is stated together with its limits in Section 11, pp. 41–42.
Results				
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram (see fig 1).	Section 2.1, p. 5; Figure 1, p. 7; Supplementary Material, Section S2 and Table S1	In total, 1 333 records, 296 duplicates, 340 reports assessed in full text, 138 excluded, and 202 studies included.
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Section 2.2, p. 7; Supplementary Material, Section S4; data repository	Three reports without a held-out set are cited with their exclusion reasons in the paper; exclusion reasons for all 138 reports, with quoted evidence, in the repository.
Study characteristics	17	Cite each included study and present its characteristics.	Table 8, p. 22; Table 11, p. 30; Table 12, p. 32; Table 13, p. 33; data repository	The paper presents 38 representative studies; the evidence table of the repository gives the characteristics of all 202 studies.
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 16, p. 38; Table 17, p. 38; data repository	Summary by task in Table 16; 38 representative studies in Table 17; labels and signs for each of the 202 studies in the repository.
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Tables 8, 11, 12, 13; Table 9, p. 26; Table 10, p. 28; Figure 23, p. 35; data repository	Primary metric of each study; external-validation pairs of values for 13 studies.
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Sections 6.1–6.5.3, pp. 20–35	Characteristics and protocols of the studies in each group.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Section 9.2, pp. 39–40; Section 6.5.3, p. 34	Proportions with 95% CIs; the suspected-leakage group has higher accuracy than the clean group (Mann–Whitney, Hodges–Lehmann with CI); median drop under external validation.
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section 9.2, pp. 39–40; Figure 22, p. 32; Section 6.5.3, p. 34	By data source, imaging conditions, protocol level, metric type, and type of shift.
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesised results.	Section 9.2, pp. 39–40	Van Elteren, rank regression with source effects, and Fisher, with <i>p</i> values after Holm correction; three-group comparison (exploratory).
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Section 2.2, p. 7 (Statistical methods)	Not assessed; see item 14.
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Section 2.2, p. 7 (Statistical methods)	Not assessed; see item 15.
Discussion				
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 1, p. 2; Sections 6–10; Section 12, p. 42	Interpretation by RQ1–RQ4 and comparison with prior reviews.
	23b	Discuss any limitations of the evidence included in the review.	Section 9.1, p. 39; Section 9.2, p. 39	Limitations of data, models, and protocols.
	23c	Discuss any limitations of the review processes used.	Section 11, p. 41	Open-access scope, search, retrospective registration, no agreement coefficient, leakage labels, and experimental design.

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
	23d	Discuss implications of the results for practice, policy, and future research.	Section 10.1, p. 40; Section 10.2, p. 41; Section 12, p. 42	Recommendations, research directions, and priorities.
Other information				
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Section 2, p. 5; Section 11, p. 41	Open Science Framework (OSF) Registries (code fwz4s, DOI: 10.17605/OSF.IO/FWZ4S); its retrospective nature is stated in the limitations.
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Section 2, p. 5; data repository	Protocol on OSF and in the data repository.
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Section 2, p. 5; Section 11, p. 41; Supplementary Material, Table S2; data repository	The limitations state the retrospective registration (2026-10-03), where it differs from the conduct of the review (Supplementary Material, Section S3 and Table S3), and how the scope was narrowed; the list of amendments with rationale and consequences is in Table S2 of the Supplementary Material; the full protocol with amendments 1–12 is in the repository.
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Declarations, p. 43	No external funding was received.
Competing interests	26	Declare any competing interests of review authors.	Declarations, p. 43	No competing interests.
Availability of data, code, and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data Availability Statement, p. 43; data repository	Publicly available on Zenodo (CC BY 4.0): coding guide and forms, extracted data, data used for the analyses, run commands, and per-run results of the experiment. Analysis code is not released but is available upon request.

PRISMA 2020 for Abstracts checklist

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
Title				
Title	1	Identify the report as a systematic review.	Title	Yes.
Background				
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Abstract, sentences 1–2	Gap and objectives: systematizing tasks, models, and datasets, and appraising protocols.
Methods				
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Abstract, sentences 3 and 5	English-language articles from 2016–2026 in fully open-access journals that train deep learning models on RGB images of rice or maize leaves; reports with clear leakage excluded.
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Abstract, sentence 3	Scopus, IEEE Xplore, ScienceDirect, MDPI; searched on 2026-09-21–22 (the abstract gives September 2026).
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Abstract, sentences 4–5	Six leakage signs coded at full text; reports with clear leakage excluded.
Synthesis of results	6	Specify the methods used to present and synthesise results.	Abstract, sentence 4	Coding by task, architecture family, dataset, and protocol; controlled experiment.
Results				

Topic	Item	Checklist item (verbatim PRISMA 2020)	Location in the manuscript	Comment
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Abstract, sentences 5–7	In total, 202 studies; distribution by task, year, architecture family, and dataset; no human participants.
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Abstract, sentences 8–10	Leakage rates, external-validation drop, experimental results with the direction of the effect (the leakage arm is 1.2–2.8 percentage points higher); the comparison of the suspected-leakage and clean groups is presented only in the paper.
Discussion				
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Abstract, sentences 7–9	Evidence concentrated on few datasets, leakage, no separate validation set, and lack of external validation.
Interpretation	10	Provide a general interpretation of the results and important implications.	Abstract, sentence 11	Interpretation and recommendations.
Other				
Funding	11	Specify the primary source of funding for the review.	Not stated in the abstract; Declarations	No external funding was received.
Registration	12	Provide the register name and registration number.	Not stated in the abstract; Section 2, p. 5	OSF Registries (DOI: 10.17605/OSF.IO/FWZ4S), retrospective registration.

References

- [1] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, Roger Chou, Julie Glanville, Jeremy M. Grimshaw, Asbjørn Hróbjartsson, Manoj M. Lalu, Tianjing Li, Elizabeth W. Loder, Evan Mayo-Wilson, Steve McDonald, Luke A. McGuinness, Lesley A. Stewart, James Thomas, Andrea C. Tricco, Vivian A. Welch, Penny Whiting, and David Moher. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021. doi: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71).
- [2] Prabira Kumar Sethy. Rice leaf disease image samples. Mendeley Data, 2024. Version 2, published 18 July 2024.